PHY1203 - Modern Physics
2025 - Lesson 06

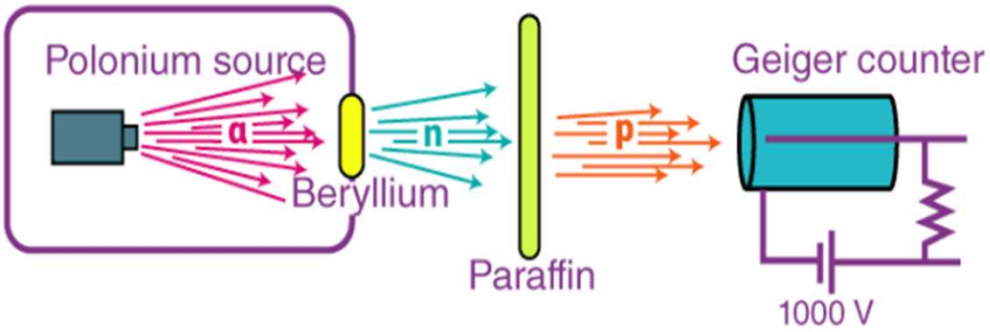


The discovery of neutron

The neutron was discovered by Sir James Chadwick in 1932.

He was awarded the Nobel Prize in Physics in 1935 for his groundbreaking discovery.





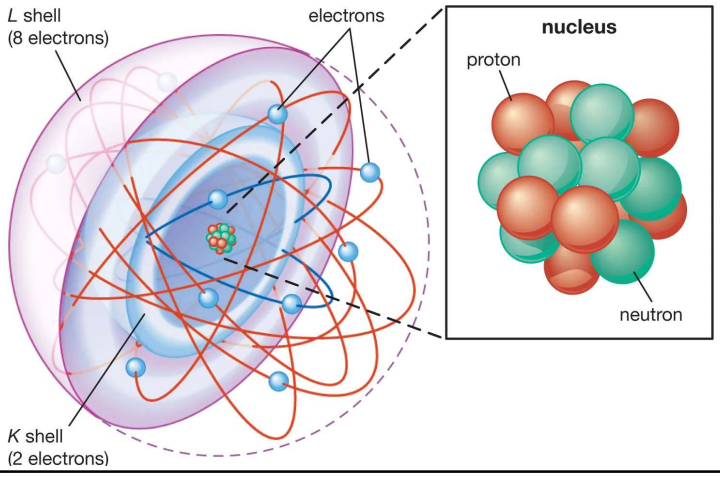
He bombarded beryllium with alpha particles and observed neutral particles being emitted.

His experiments, which proved the existence of this chargeless subatomic particle, explained the differences in atomic mass and were important for understanding the atom's structure.

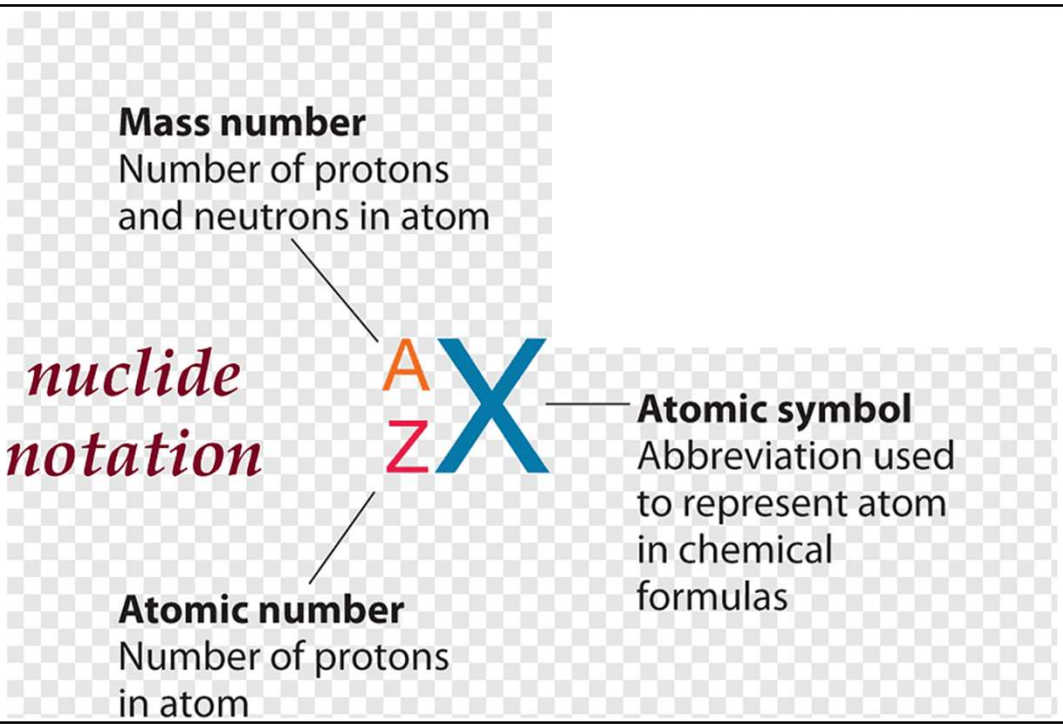
Atomic Nucleus

All atomic nuclei are composed of protons and neutrons. Exception is the hydrogen nucleus, which has a proton only.

Nucleon is the term used to refer to either a proton or a neutron.



- **Atomic number Z, which equals the number of protons in the nucleus (also called proton number)**
- **Mass number A, which equals the number of nucleons in the nucleus (also called nucleon number)**
- **Neutron number N, which equals the number of neutrons in the nucleus**



The nuclei of all atoms of one element must contain the **same number of protons.** But for some elements the number of neutrons may be different.

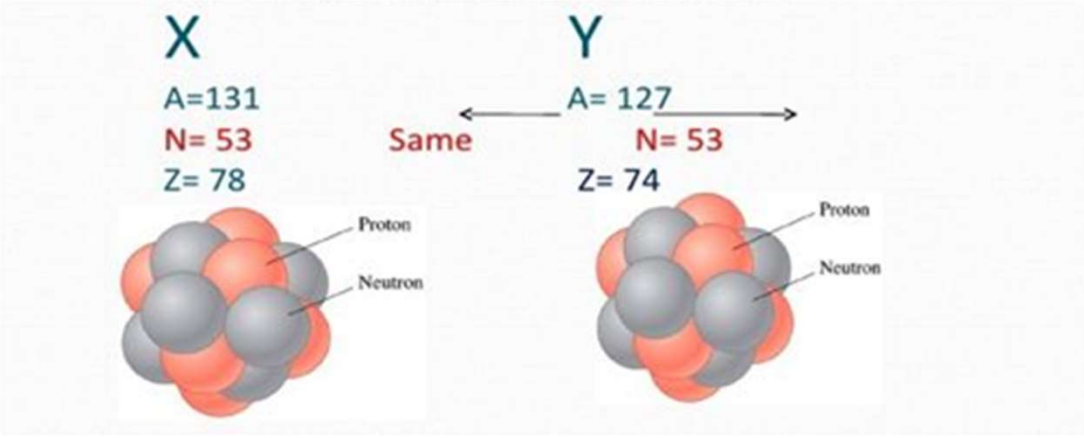
Isotopes

Nuclei with same atomic number and different mass number (different number of neutrons) are are called **isotopes.**

e.g.
- Hydrogen: ${}_{1}^{1}\mathrm{H}$, ${}_{1}^{2}\mathrm{H}$, ${}_{1}^{3}\mathrm{H}$ or ${}_{1}^{1}\mathrm{H}$, ${}_{1}^{2}\mathrm{D}$ (deuterium), ${}_{1}^{3}\mathrm{T}$ (tritium)
- Carbon : ${}_{6}^{12}\mathrm{C}$, ${}_{6}^{13}\mathrm{C}$, ${}_{6}^{14}\mathrm{C}$
- Magnesium : ${}_{12}^{24}\mathrm{Mg}$, ${}_{12}^{25}\mathrm{Mg}$, ${}_{12}^{26}\mathrm{Mg}$
- Calsium : ${}_{20}^{40}\mathrm{Ca}$, ${}_{20}^{41}\mathrm{Ca}$, ${}_{20}^{42}\mathrm{Ca}$,……${}_{20}^{48}\mathrm{Ca}$ (9 isotops)

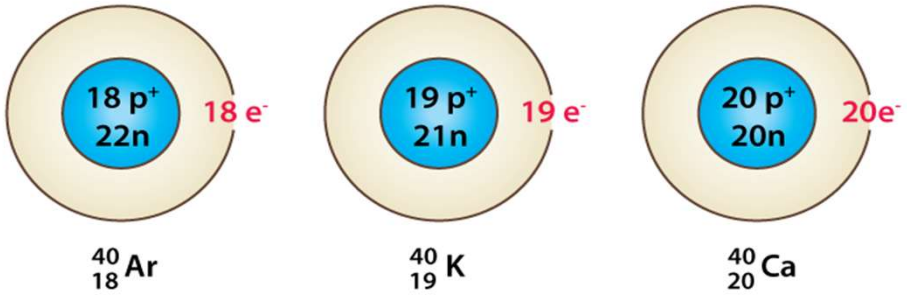
Isotones

Atoms having the same number of neutrons are called isotones.

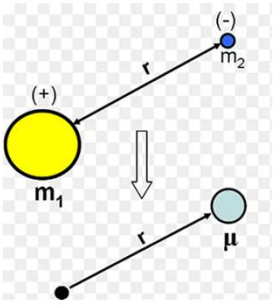


Isobars

They are different elements with same atomic mass .



Isotope shifts and deuterium spectrum

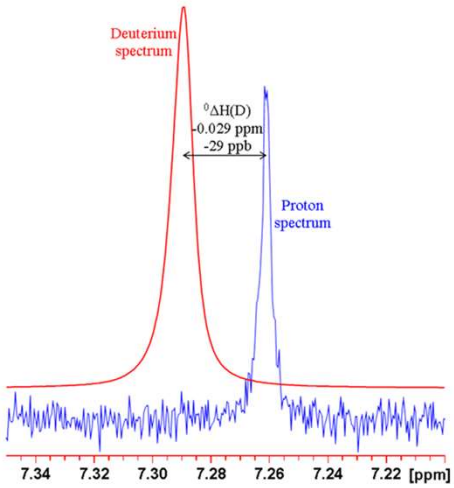


$$\mu = \frac{1}{\frac{1}{m_1} + \frac{1}{m_2}} = \frac{m_1 m_2}{m_1 + m_2},$$

$$\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$ where $$R_H = \frac{me^4}{8h^3 c\varepsilon_0^2}$$ is called the Rydberg constant.

$R_H = 1.09677576x10^7\, m^{-1}$ for hydrogen.

$R_\infty = 1.0973731x10^7\, m^{-1}$ in the limiting case for heavy elements



Charge and Mass

The Sub-atomic Particles			
Relative size	Name	Mass (Kg)	Charge (C)
	Proton	1.67×10^{-27}	$+1.602 \times 10^{-19}$
	Neutron	1.67×10^{-27}	0
 size exaggerated	Electron	9.11×10^{-31}	-1.602×10^{-19}

Particle	Mass	
	kg	Relative
Proton (m_p)	1.673×10^{-27}	1 836.15
Neutron (m_n)	1.675×10^{-27}	1 838.68
Electron (m_e)	9.109×10^{-31}	1

Unified Mass Unit (Atomic Mass Unit – amu)

Unified mass unit, $u \equiv \frac{1}{12}$ the mass of a neutral C^{12} atom

by definition (of a mole) 1 mole, C^{12} = 12 grams

$$1\ \text{u} = \frac{10^{-3}\ \text{kg}}{6.022 \times 10^{23}} = 1.66 \times 10^{-27}\ \text{kg}$$

Electron volt

An electron volt (eV) is a unit of energy equal to the amount of kinetic energy an electron gains when it moves through an electric potential difference of one volt.

$$1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$$

$$1\ \text{MeV} = 1.602 \times 10^{-13}\ \text{J}$$

Standard Values

Particle	Mass		
	kg	u	MeV/c^2
Proton (p)	1.6726×10^{-27}	1.007276	938.28
Neutron (n)	1.6750×10^{-27}	1.008665	939.57
Electron (e)	9.11×10^{-31}	5.5×10^{-4}	0.511

Energy of 1 u = 931.5 MeV

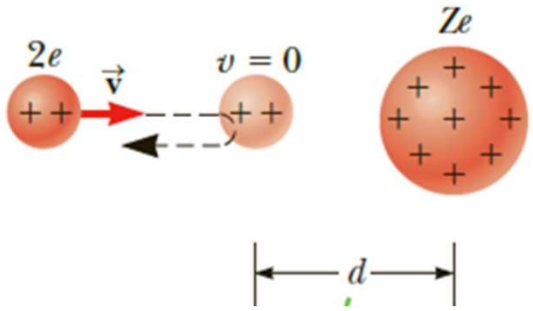
Mass of 1 u = 931.5 MeV/c^2

Don't confuse the atomic mass the mass number !

e.g: For sodium Na
atomics number = 23 mass number = 22.99

Size of the Nuclei

The size of the nuclei was investigated by Rutherford using the principle of conservation of energy. He obtain an expression for the closest distance that a positive (+2e) alpha particle can reach a positive nucleus (+eZ) in head-on collision.



$$\frac{1}{2}mv^2 = \left(\frac{1}{4\pi\epsilon_o}\right)\frac{(2e)(Ze)}{d}$$

$$d = \frac{4Ze^2}{4\pi\epsilon_o mv^2}$$

From the experiment, he found that α-particles approached a distance $d = 3.2\times10^{-14}$ m to the nucleus. That indicated the radius of the gold nucleus must be less that that value. For Ag $d = 2\times10^{-14}$ m.

From this results Rutherford concluded that

- the positive charge in an atom is concentrated in a small sphere which he called the nucleus and
- that the radius of the nucleus is less than 10^{-14} m.

Other experiments also have shown the nucleus is spherical and has an average radius given by

$$r = r_o A^{1/3}$$

where $r_o = 1.2 \times 10^{-15}$ m is a constant and A is the number of nucleons.

For nuclear sizes measurements are given by fm = 10^{-15} m.

Size of the Nuclei

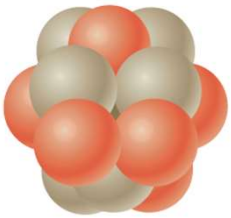


Figure 29.2 A nucleus can be visualized as a cluster of tightly packed spheres, each of which is a nucleon.

- Large Electrostatic force (Coulomb force) is a repulsive force between positive protons
- Short range (2 fm) Nuclear Force (strong force) is an attractive force between nucleons.

The protons attract each other via the nuclear force, and at the same time they repel each other through the Coulomb force.

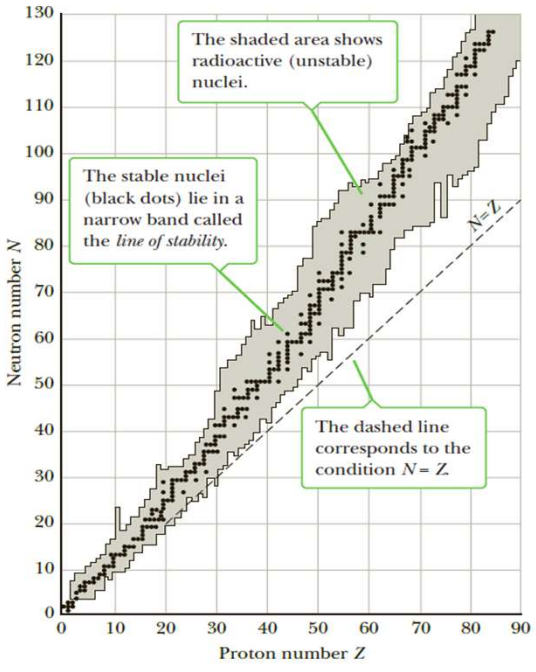
The attractive nuclear force also acts between pairs of neutrons (n-n) and between neutrons and protons (n-p).

Nuclear Stability

There are about 260 stable nuclei; hundreds of others have been observed but are unstable.

light nuclei are most stable if they contain equal numbers of protons and Neutrons ($N = Z$)

Heavy nuclei are more stable if $N > Z$



This difference can be partially understood by recognizing that as the number of protons increases, the strength of the Coulomb force increases, which tends to break the nucleus apart.

As a result, more neutrons are needed to keep the nucleus stable because neutrons are affected only by the attractive nuclear forces.

The extra neutrons reduce the nuclear charge density.

Binding Energy (*B*)

- The total mass of a nucleus is always less than the sum of the masses of its nucleons!
  i.e. the total energy of the bound system (the nucleus) is less than the combined energy of the separated nucleons

- This difference in energy is called the **binding energy of the nucleus** and can be thought of as the energy that must be added to a nucleus to break it apart into its separated neutrons and protons

E.g.: Calculate the binding energy of deuteron in MeV. The mass of the deuterium nucleus is m_D = 2.014 103 u.

m_p = 1.007 825 u and m_n = 1.008 665 u

$$\Delta m = (m_p + m_n) - m_D$$

$$= (1.007\ 825\ \text{u} + 1.008\ 665\ \text{u}) - 2.014\ 103\ \text{u}$$

$$\Delta m = 0.002\ 388\ u$$

$$= 0.002\ 388 \times 931.5\ \text{MeV}/c^2$$

$$B = 2.224\ \text{MeV}$$

Problem (1): Calculate the binding energy of alpha particles.

m_p = 1.007 825 u m_n = 1.008 665 u

m_{He} (4,2)= 4.00260 u u = 931.5 MeV/c^2

$$\Delta m = (2m_p + 2m_n) - m_{He}$$

$$= [(2 \times 1.007\ 825 + 2 \times 1.008\ 665) - 4.00260]\ \text{u}$$

$$\Delta m = 0.03\ 0038\ u$$

$$B = 0.03\ 0038 \times 931.5\ \text{MeV}/c^2$$

$$B = 28.30\ \text{MeV}$$

$$\bar{B} = \frac{B}{A} = \frac{28.30}{4} = 7.07\ \text{MeV}$$

Problem (2): Calculate the binding energy of $^{56}_{26}\mathrm{Fe}$ nucleus.

$m_p = 1.007\ 825$ u $\qquad m_n = 1.008\ 665$ u

$m(^{56}_{26}\mathrm{Fe}) = 55.934\ 939$ u $\qquad u = 931.5\ \frac{\mathrm{MeV}}{c^2}$

$$\Delta m = (26m_p + 30m_n) - m(^{56}_{26}\mathrm{Fe})$$

$$= [(26 \times 1.007\ 825 + 30 \times 1.008\ 665) - 55.934\ 939]\mathrm{u}$$

$$\Delta m = 0.52\ 8461\ u$$

$$= 0.52\ 8461 \times 931.5\ \mathrm{MeV}/c^2$$

$$B = 492.26\ \mathrm{MeV}$$

$$\bar{B} = \frac{B}{A} = \frac{492.26}{56} = 8.79\ \mathrm{MeV}$$

